

UMET9U-30-1 Statistics Portfolio — Chat Notes

Part 1: Assignment Summary

Core details at a glance

- **Module:** UMET9U-30-1 — Professional Skills for Banking and Finance
- **Component:** Portfolio — Statistics component
- **Weighting:** 33% of the total module mark
- **Submission deadline:** Before **14:00 on 12/05/2026** (UK local time, 24-hour clock)
- **Late submission:** Eligible for the 48-hour late submission window
- **Marking/feedback returned by:** 11/06/2026
- **Word limit:** 1,500 words (includes headings, citations, quotes, lists; **excludes** statistical tables and reference list)
- **Format required:** Microsoft Word (.docx), 12-point Times New Roman or Arial
- **Submission method:** Electronic submission via Blackboard only (no email/paper/disk)

The topic

The assignment is built around the research question studied in Mauro (2003): **can lagged (previous-year) stock returns predict next-year economic growth, and does this relationship differ between emerging and developed economies?**

The economic reasoning is that financial markets react quickly to new information while macroeconomic variables react more slowly, so last year's stock returns may act as a leading indicator for this year's output growth.

Two variables are used:

- **r — Stock returns:** Bloomberg stock market indices converted to log-returns, minus inflation (consumer prices, FRED).
- **Dy — GDP growth:** Constant GDP per capita (FRED, 2010 USD) converted to log-returns.

Where to find the data

The brief states: *"using data provided on Blackboard"*. The dataset covers **10 emerging + 10 developed economies**, classified using the MSCI Market Classification.

You must pick ONE emerging economy and ONE developed economy and use only those two countries for your analysis. The choices are:

- **Emerging:** Mexico (MEXBOL), Brazil (IBOV), Poland (WIG20), Hungary (BUX), Turkey (XU100), South Africa (TOP40), China (SHCOMP), India (NIFTY), Malaysia (FBMKLCI), Korea (KOSPI).
- **Developed:** Canada (SPTSX), USA (SPX), France (CAC), Germany (DAX), Italy (FTSEMIB), Switzerland (SMI), UK (UKX), Australia (AS51), Japan (NKY), New Zealand (NZSE50FF).

Where to find the template

The brief does **not** reference a named template file. It only specifies the formatting rules (see "Formatting" below) and says the first page must contain your student number, module name/number, word count, and coursework title. If a starter document exists, it would be on Blackboard alongside the data — check the Coursework tab there, but plan to build the document yourself if nothing is posted.

What you actually need to produce

A statistics report ($\leq 1,500$ words) with the following sections. Each of these six bullets maps roughly to a section of your report:

1. **Short introduction** to the relationship between stock returns and economic growth (cite Mauro 2003 and related literature).
2. **Descriptive statistics** for both stock returns and GDP growth — compute them and *discuss* what they say (mean, SD, skewness, kurtosis, min/max, etc.).
3. **Distributional analysis** using a **histogram** and a **box plot** for each variable; comment on shape, outliers, spread.
4. **Relationship analysis** between lagged stock returns (year $t-1$) and economic growth (year t) using a **correlation coefficient** and a **scatter plot**.
5. **Regression model:** GDP growth (dependent) on lagged stock returns (explanatory).
 - Interpret the slope coefficient.
 - Perform a **t-test** for statistical significance.
 - Discuss R^2 .
 - Then run a **second regression** that also includes a **lagged dependent variable** as a robustness check.
6. **Short conclusion** summarising findings and comparing the emerging vs developed economy results.

All numerical work is expected to be done in **Excel** — the marking rubric explicitly rewards Excel-based descriptive statistics, regression output, and graphics.

How the work is marked

Criterion	Weight
Ability to compute descriptive statistics, regression output and produce graphics using Excel	40%
Ability to correctly interpret statistical results	40%
Writing and presentation	20%

Markers will specifically look for: a structured report with intro/conclusion; suitable descriptive statistics *and* their discussion; suitable graphics *and* their discussion; interpretation of correlation and scatter; interpretation of regression coefficients, t-tests and R^2 ; comments on the research question; references; and acknowledgement of **weaknesses or possible improvements** of your analysis (easy to forget — include it).

Practical workflow suggestion

A reasonable order of attack:

1. Pick your two economies (e.g. one you have some familiarity with in each group).
2. Download/extract the Blackboard dataset into Excel and isolate the two country columns.
3. Build a working sheet: compute descriptives (use the Data Analysis ToolPak → Descriptive Statistics).
4. Generate the histograms and box plots, then the scatter with lagged returns on the x-axis.
5. Compute the correlation, then run the two regressions (Data Analysis → Regression). Note that the second regression needs a lagged GDP growth column as an additional explanatory variable.
6. Write up the report in Word around the Excel outputs. Paste tables/charts as images or native objects.
7. Reference using **UWE Harvard**.

Formatting and first-page checklist

First page must show:

- Student number
- Module name and number (Professional Skills for Banking and Finance / UMET9U-30-1)
- Word count
- Coursework title

Body text: 12-point Times New Roman or Arial, Word format.

Important rules to keep in mind

- **No generative AI** for producing the content of the report (analysis, discussion, etc.) — UWE policy on AI use applies.
 - Referencing must use **UWE Harvard**.
 - Submit via Blackboard only; submit early to avoid deadline congestion; keep the receipt.
 - The module leader **cannot** grant extensions — for Personal Circumstances, contact a Student Support Adviser.
 - Learning outcomes assessed: **MO1-MO3** (see the module specification linked in the brief).
-

Part 2: A Worked Example — How I'd Approach It

One upfront point: the brief prohibits using generative AI to *produce* the content of your report. So the example below shows **the methodology** with illustrative numbers — think of it as a worked example to learn from and then redo with your real data and your own words. The numbers are plausible but **made up** for demonstration; your actual Excel output will differ.

For the example I'll pick **India** (emerging) and **Germany** (developed) — two economies with very different growth regimes, which makes for a more interesting comparison.

Step 1 — Set up the Excel workbook

Open the Blackboard dataset and on a clean sheet, set up one block per country with these columns:

Year	r (real stock return)	Dy (GDP growth)	r(-1) lagged	Dy(-1) lagged
1990	0.14	0.045	—	—
1991	-0.08	0.012	0.14	0.045
1992	0.22	0.058	-0.08	0.012
...	...	...	...	...

The lagged columns are the key construction: **$r(-1)$ in year $t = r$ in year $t-1$** . In Excel, if r is in $B2:B31$, the lagged column in C starts at C3 with $=B2$ and drags down. The first year has no lag so it's dropped from the regression sample.

Do this for both countries on separate sheets (**India**, **Germany**) so your analyses don't get tangled.

Step 2 — Descriptive statistics

Use **Data → Data Analysis → Descriptive Statistics** (enable the Analysis ToolPak in Excel Options first if it isn't on). Tick "Summary statistics". Run it for each of the four series (r and Dy for each country).

Illustrative output:

	India r	India Dy	Germany r	Germany Dy
Mean	0.061	0.048	0.041	0.015
Median	0.075	0.052	0.058	0.018
Std. dev.	0.221	0.027	0.183	0.021
Min	-0.42	-0.06	-0.45	-0.05
Max	0.51	0.089	0.38	0.041
Skewness	-0.35	-0.88	-0.52	-1.10
Kurtosis	0.81	1.45	1.22	2.10
Obs	30	30	30	30

What to discuss (this is where marks are won):

- India's stock returns have a higher mean (6.1% vs 4.1%) *and* higher volatility (SD 22% vs 18%) — consistent with the emerging-markets "higher risk, higher return" profile.
- India's GDP growth averages ~4.8% vs Germany's ~1.5%, reflecting catch-up growth.
- Both Dy series are negatively skewed with excess kurtosis — recessions are rare but deep (useful framing for the histogram section).

Step 3 — Histogram and box plot

For each of the four series:

- **Histogram:** Data Analysis → Histogram. Define bins (e.g. 8-10 bins covering the range). Tick "Chart Output".
- **Box plot:** Insert → Chart → Box & Whisker (Excel 2016+).

Example discussion lines you'd write:

Germany's GDP growth histogram shows a concentrated distribution around 1.5-2.5%, with a long left tail driven by the 2009 and 2020 downturns. The box plot confirms this asymmetry:

the whisker extends further below the box than above, and two observations fall outside the $1.5 \times \text{IQR}$ fences.

India's stock returns are more symmetric but with fatter tails than a normal distribution would predict, visible as a flatter histogram peak and longer whiskers on the box plot.

Step 4 — Correlation + scatter

Compute the correlation between $r(-1)$ and Dy :

```
=CORREL(C3:C31, D3:D31)
```

Illustrative results:

- India:** $\text{corr}(r(-1), Dy) = 0.24$
- Germany:** $\text{corr}(r(-1), Dy) = 0.42$

Then **Insert → Scatter** with $r(-1)$ on the x-axis and Dy on the y-axis. Right-click a point → "Add Trendline" → Linear, and tick "Display Equation" and "Display R-squared".

Example interpretation:

The positive correlation in both countries is consistent with Mauro's (2003) hypothesis that stock returns lead output growth. The relationship is visibly tighter for Germany — points cluster closer to the trend line — while India's scatter is noisier, suggesting that Indian GDP is driven by factors less captured by its stock market (agriculture, informal sector, policy shocks).

Step 5 — Regression 1

Data Analysis → Regression. Input Y range: Dy . Input X range: $r(-1)$. Tick "Labels".

Illustrative output for Germany:

	Coefficient	Std. Error	t-stat	p-value
Intercept	0.012	0.004	3.00	0.006
$r(-1)$	0.045	0.019	2.37	0.025

$R^2 = 0.16$, Observations = 29.

How to interpret (this is the meat of the report):

The estimated slope of 0.045 means that a 1 percentage point increase in the previous year's real stock return is associated with an additional 0.045 percentage points of GDP growth the following year. The t-statistic of 2.37 exceeds the 5% critical value of ~ 2.05 ($df = 27$), and the p-

value of 0.025 confirms rejection of $H_0: \beta = 0$. Stock returns are therefore a statistically significant leading indicator of German GDP growth.

The R^2 of 0.16 indicates that 16% of the variation in German GDP growth is explained by lagged stock returns alone — economically meaningful but modest, as expected given the many other determinants of output.

For India, you'd likely find something like $\beta = 0.028$, $t \approx 1.2$, $p \approx 0.24$, $R^2 = 0.06$ — not significant. That asymmetry is the interesting story.

Step 6 — Regression 2 (robustness with lagged Dy)

Same procedure but the X range now covers **two columns**: $r(-1)$ and $Dy(-1)$. Select them together (make sure they're adjacent on the sheet, or copy them side-by-side).

Illustrative output for Germany:

	Coefficient	Std. Error	t-stat	p-value
Intercept	0.008	0.004	2.00	0.056
$r(-1)$	0.038	0.018	2.11	0.044
$Dy(-1)$	0.320	0.175	1.83	0.079

$R^2 = 0.26$.

Why this matters, in words for your report:

Adding lagged GDP growth controls for persistence in the output series — the fact that a good year tends to be followed by another good year. The coefficient on $r(-1)$ falls slightly from 0.045 to 0.038 but remains statistically significant, indicating that stock returns have predictive power over and above the autoregressive component of GDP. The R^2 rises to 0.26, and the lagged dependent variable itself is weakly significant ($p = 0.079$). This robustness check strengthens the case that stock returns carry genuine leading information about the real economy.

Step 7 — Structure of the 1,500-word write-up

A rough word budget that fits the brief:

- Introduction (~250 words):** the leading-indicator hypothesis, cite Mauro (2003), state that you'll compare one emerging and one developed economy.
- Data and descriptive statistics (~300 words):** describe the variables, the two countries, and the table. Draw out the mean/SD/skewness contrast.
- Distributional analysis (~250 words):** embed the histograms and box plots (these don't count to word limit as statistical outputs, but your discussion does). Discuss tails, outliers,

asymmetry.

- **Correlation and scatter (~200 words):** report the two correlations, describe the scatters, note the difference.
- **Regression analysis (~350 words):** both regressions, with the interpretation template above. This is the highest-mark section so give it room.
- **Conclusion (~150 words):** the comparative finding (stock returns as leading indicator: strong for Germany, weak for India), plus **weaknesses/improvements** — small sample (30 years), only one country per group so limited external validity, potential structural breaks (e.g. 2008, 2020), no controls for monetary policy or commodity cycles.
- **References:** Mauro (2003), plus 2-4 others (e.g. Fama 1981 on stock returns and real activity, Stock & Watson on leading indicators).

Step 8 — Polish before submission

- Front page with student number, module name/code, word count, title.
- Body in 12-point Times New Roman or Arial.
- All tables pasted as editable Excel objects or high-resolution images, with captions ("Table 1: Descriptive statistics, 1995-2024").
- UWE Harvard references.
- Run your own word count on the body (excluding tables and references).